Amir Adam

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EDUCATION AND TRAINING

16/08/2022 – CURRENT Mohali , India

BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATIONS Chandigarh University

Relevant focus: Embedded Systems, VLSI, Communications

Website https://www.cuchd.in/ | **Level in EQF** EQF level 6

13/02/2017 – 01/12/2020 Wau, South Sudan

CERTIFICATE OF SECONDARY EDUCATION Loyola Secondary School

General Sciences: Maths,Physics,Chemistry,Biology

Level in EQF EQF level 4

SKILLS

Programming

C/C++ | Python | Typescript

Embedded Systems

Microcontrollers: ESP32,STM32 | Raspberry Pi | Sensor Integration

Tools & Platforms

Git | STM32CUBEIDE | Arduino IDE | VSCode | GitHub

App Development

React Native | Supabase | Expo/Expo router | PostgreSQL

Conceptual Knowledge

Communication Protocols: I2C,SPI,UART | Embedded Systems Architecture | Networking Basics(HTTP, client-server side) | Basic PCB Design

LANGUAGE SKILLS

Mother tongue(s): **ARABIC(SUDANESE DIALECT)**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PROJECTS

20/10/2025 – 20/11/2025

Disaster-Resilient Offline Mesh Communication System

Technologies: ESP32, LoRa

- Designed and developed an offline mesh communication system using ESP32 microcontrollers and LoRa to enable connectivity in network-isolated environments.
- Implemented peer-to-peer mesh networking architecture to support decentralized message transmission without reliance on cellular or internet infrastructure.
- Integrated sensor data acquisition and local processing to enhance situational awareness in emergency scenarios.
- Optimized embedded firmware for low-power consumption, improving system reliability in constrained and disaster-prone environments.

Link https://techtacklenerd.blogspot.com/2025/11/disaster-resilient-hybrid-mesh.html

15/02/2026 – 19/02/2026

Full-Stack IoT Monitoring System with Web Dashboard

Technologies: ESP32, React(Vite), Python (Flask), HTTP, REST API

- Designed and developed a full-stack IoT system for real-time sensor data acquisition and visualization.
- Implemented firmware on ESP32 to collect environmental sensor data and transmit it to a backend server via HTTP requests.
- Built a RESTful API using Flask to handle data ingestion, processing, and storage.
- Developed a responsive frontend dashboard using React.js to display real-time and historical sensor data.
- Enabled dynamic data updates and visualization through API integration between frontend and backend.
- Improved system reliability by handling network delays and ensuring consistent data transmission.

Link https://github.com/amirphiladam2/FullStack-IoT

20/10/2024 – 22/10/2024

Home Automation System using ESP32 and PySerial

Technologies: ESP32, Python (PySerial)

- Designed a serial communication-based home automation system for controlling electrical appliances.
- Programmed ESP32 firmware to receive and execute control commands via UART communication.
- Integrated relay modules to enable switching of AC loads (lamps) based on microcontroller signals.
- Developed a Python-based interface using PySerial to transmit real-time control commands from a computer to ESP32.

Link https://techtacklenerd.blogspot.com/2024/10/home-automation-project-using-esp32-and.html

CREATIVE WORKS

12/07/2023 – CURRENT

YouTube Tech Content Creator

- Create technical educational content focused on electronics and app development concepts.
- Produce video tutorials demonstrating real-world implementations using STM32,ESP32 and React Native.

Link https://www.youtube.com/@amirdevstudios/videos

HOBBIES AND INTERESTS

Hobbies

- Nature photography, with a focus on landscapes and environmental elements
- Coding and Electronics Tinkering